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NEXT-GENERATION DATA STORAGE MARKET WITH COVID-19 IMPACT ANALYSIS

GLOBAL FORECAST TO 2025

BY STORAGE MEDIUM, STORAGE SYSTEM,STORAGE ARCHITECTURE, END USERS (ENTERPRISES, GOVERNMENT BODIES, TELECOM, CLOUD SERVICE PROVIDERS), AND GEOGRAPHY

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1 EXECUTIVE SUMMARY

The COVID-19 pandemic has disrupted business activities globally. Due to the globally imposed lockdown, the enterprise sector has seen a huge loss in terms of production, with major impact witnessed by the manufacturing, BFSI, and retail sectors. The lockdown has severely impacted the manufacturing companies as almost every plant has come to a halt. Many essential commodity manufacturing companies are also unable to continue with production due to the lack of workforce amidst the pandemic. Thus, there is a slight decline in the next-generation data storage market as well due to COVID-19.

Next-generation data storage includes computing technologies and devices that are used to store, port, and extract data more quickly and efficiently. It deals with the security and management of data, and harnessing of the information to meet the application and software networking needs of end users. Next-generation data storage comprises various computer-controlled systems and equipment which are used to store and accurately retrieve data for reuse. There is a rapid increase in the volume of data generated in organizations and industries, which leads to the rising need for storage devices and technologies.

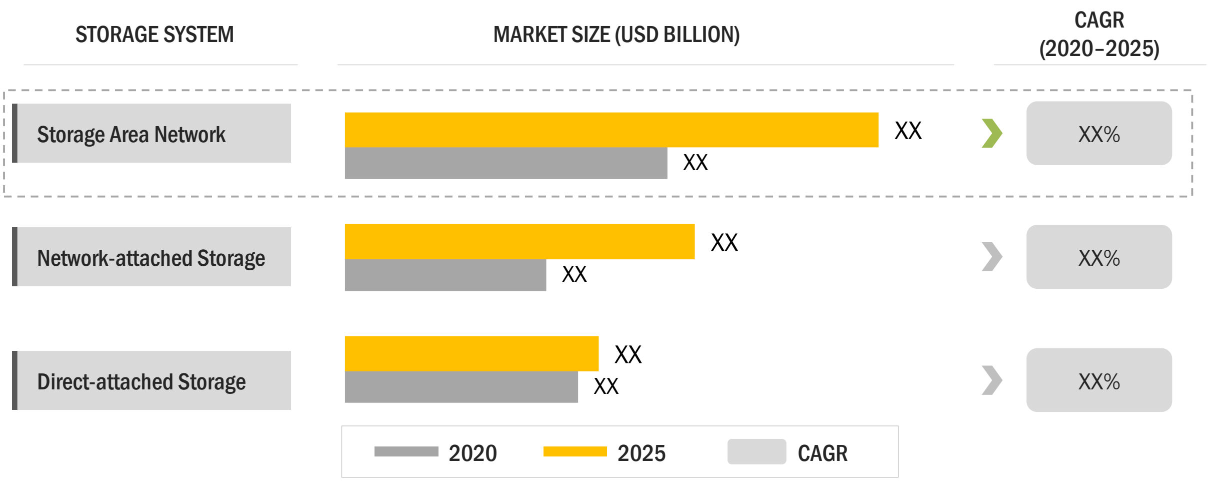
The next-generation data storage market is estimated to grow from USD XX billion by 2020 to USD XX billion by 2025; it is projected to grow at a CAGR of XX% during the forecast period. The major drivers for this market include the massive growth in digital data volume; proliferated use of smartphones, laptops, and tablets; growth of the IoT market; and increasing penetration of high-end cloud computing. The increasing number of people working from home during the pandemic is also contributing to the adoption of next-generation data storage systems by enterprises. However, breach in data on the cloud and servers, and less structured data are the major restraints for the growth of the next-generation data storage market. Also, major challenges faced by market players include high cost associated with cloud storage, and delivering robust and high-speed data storage.

The next-generation data storage market has been segmented on the basis of storage system, storage architecture, storage medium, and end user. The market based on storage system is subsegmented into direct-attached storage (DAS), network-attached storage (NAS), and storage area network (SAN). The market based on storage architecture is segmented into file and object-based storage (FOBS), and block storage. The market based on storage medium is subsegmented into hard disk drive (HDD), solid-state drive (SSD), and tape. Based on end user, the market is subsegmented into enterprises, government bodies, telecom companies, and cloud service providers. The enterprises end user is further segmented into BFSI, media and entertainment, retail, healthcare, consumer goods, manufacturing, and others.

The competitive landscape section of the report provides the ranking of the key players in the next-generation data storage market on the basis of their revenues, product offerings, and geographic reach. This chapter also describes the key growth strategies implemented by the market players from 2016 to 2020 to expand their presence globally and increase their market shares in the overall next-generation data storage market. Product launches and developments, acquisitions, partnerships, collaborations, and agreements been the key growth strategies adopted by the leading players in the market during 2016–2020. Among these strategies, the players have widely adopted product launches and developments as a key strategy to gain a competitive advantage in the market. These strategies have helped the players to efficiently cater to different end users.

Some of the leading players in this market are Hewlett Packard Enterprise Company (US), Dell Inc. (US), NetApp, Inc. (US), Hitachi, Ltd. (Japan), International Business Machines Corporation (US), Toshiba Corporation (Japan), Pure Storage, Inc. (US), Nutanix, Inc. (US), Scality (US), Micron Technology, Inc. (US), Tintri, Inc. (US), Cloudian, Inc. (US), Drobo, Inc. (US), Quantum Corporation (US), Western Digital Corporation (US), Samsung Electronics (South Korea), Fujitsu Ltd. (Japan), VMware, Inc. (US), Nexenta Systems, Inc. (US), Netgear Inc. (US), and Inspur (China).

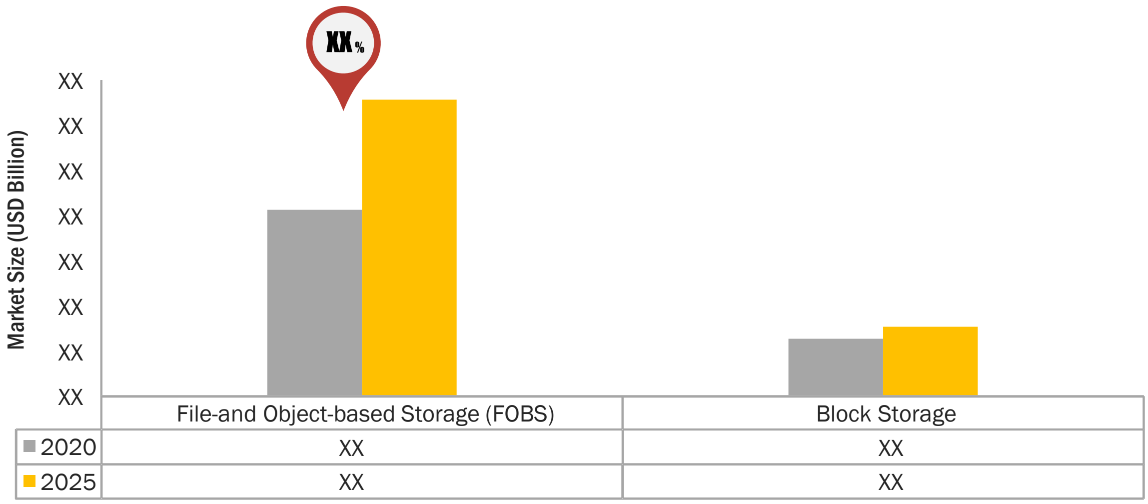
FIGURE 1 STORAGE AREA NETWORK TO HOLD LARGEST SIZE OF NEXT-GENERATION DATA STORAGE MARKET DURING FORECAST PERIOD



Source: Annual Reports, Expert Interviews, Investor Presentations, Scientific Journals, and MarketsandMarkets Analysis

The next-generation data storage market for storage area network is expected to reach USD XX billion by 2025 from USD XX billion in 2020; it is projected to grow at a CAGR of XX% during the forecast period. As an increasing number of companies have started utilizing cloud-based services to store their data with the use of virtual servers, companies aim to expand their storage capacities beyond existing infrastructural capabilities. Cloud technology creators widely use SAN technology due to its ability to connect large numbers of servers to storage devices.

FIGURE 2 FILE- AND OBJECT-BASED STORAGE TO REGISTER HIGHER CAGR IN NEXT-GENERATION DATA STORAGE MARKET DURING FORECAST PERIOD

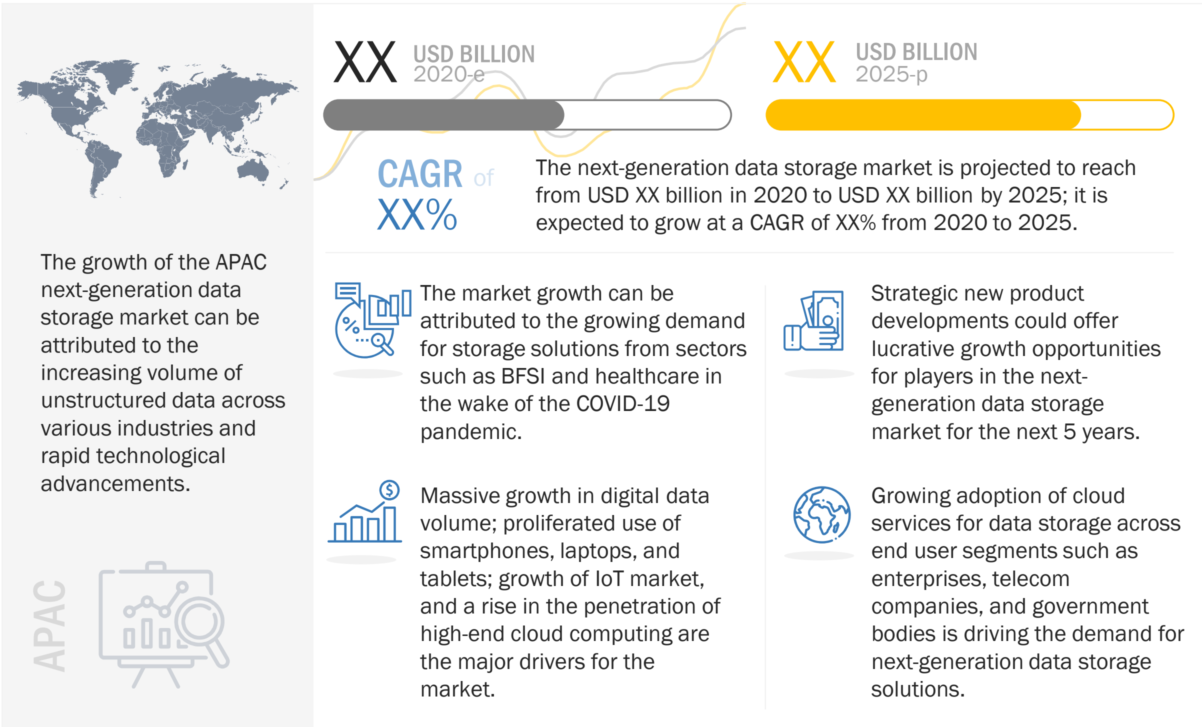


Source: Annual Reports, Expert Interviews, Investor Presentations, Scientific Journals, and MarketsandMarkets Analysis

2 PREMIUM INSIGHTS

2.1 NEXT-GENERATION DATA STORAGE MARKET OPPORTUNITIES

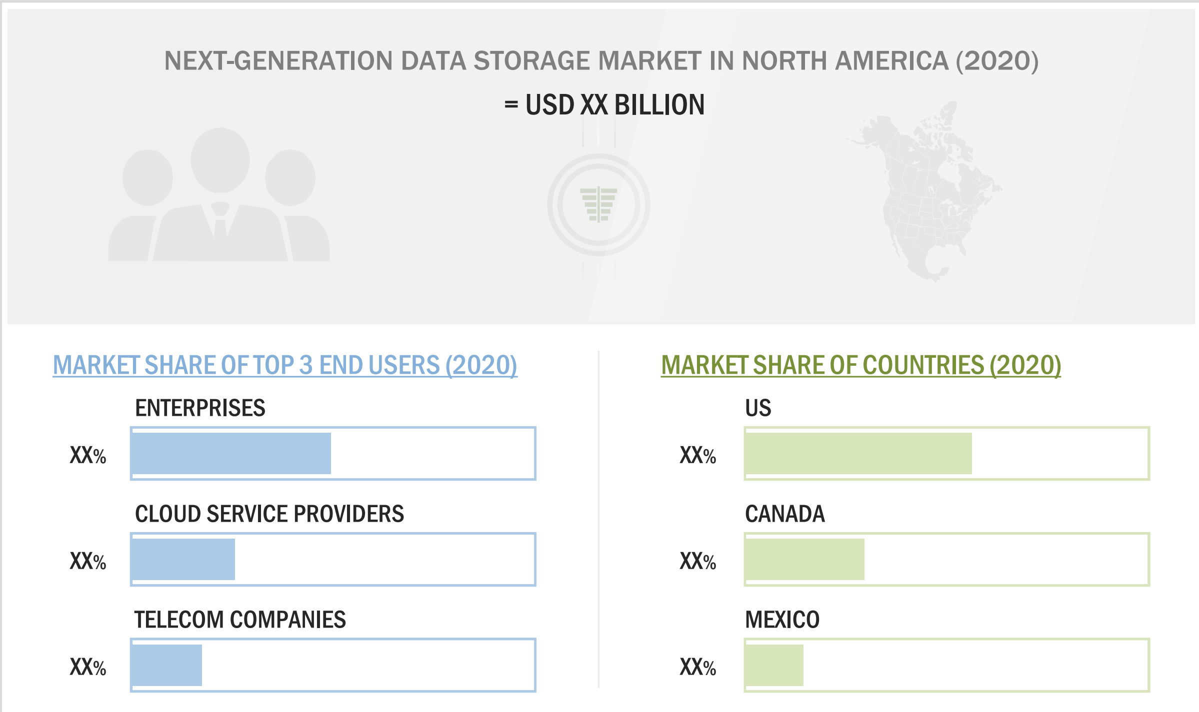
FIGURE 3 OPPORTUNITIES IN EMERGING MARKETS FOR NEXT-GENERATION DATA STORAGE TO PROVIDE LUCRATIVE GROWTH PROSPECTS DURING FORECAST PERIOD



Source: Press Releases, Investor Presentations, Scientific Journals, Expert Interviews, and MarketsandMarkets Analysis

2.2 NEXT-GENERATION DATA STORAGE MARKET IN NORTH AMERICA, BY COUNTRY AND END USER

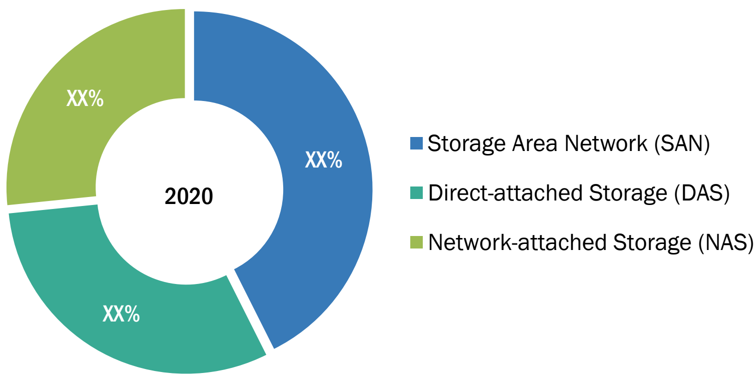
FIGURE 4 US IS EXPECTED TO HOLD LARGEST SHARE OF NEXT-GENERATION DATA STORAGE MARKET IN NORTH AMERICA IN 2020



Source: Press Releases, Investor Presentations, Scientific Journals, Expert Interviews, and MarketsandMarkets Analysis

2.3 NEXT-GENERATION DATA STORAGE MARKET, BY STORAGE SYSTEM

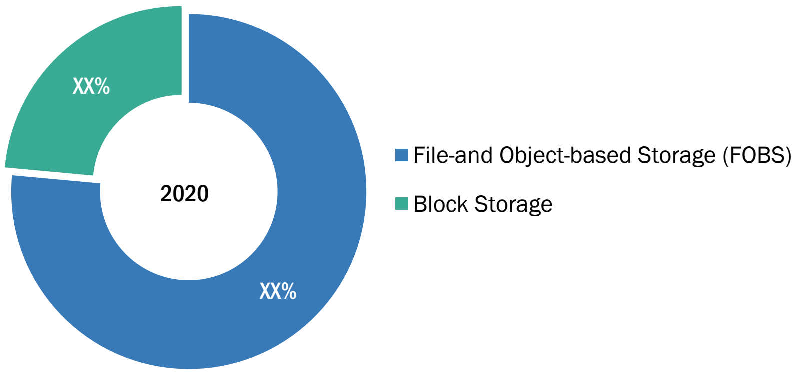
FIGURE 5 STORAGE AREA NETWORK TO HOLD LARGEST SHARE OF NEXT-GENERATION DATA STORAGE MARKET BY 2024



Source: Press Releases, Investor Presentations, Scientific Journals, Expert Interviews, and MarketsandMarkets Analysis

2.4 NEXT-GENERATION DATA STORAGE MARKET, BY STORAGE ARCHITECTURE

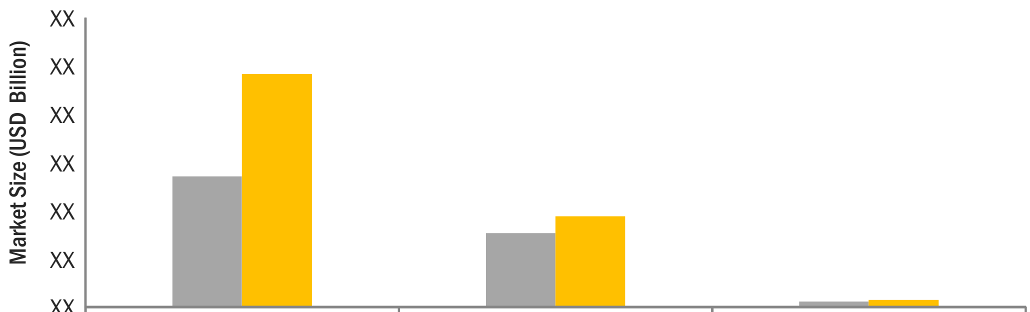
FIGURE 6 FILE- & OBJECT-BASED STORAGE TO HOLD LARGER SHARE OF NEXT-GENERATION DATA STORAGE MARKET IN 2020



Source: Press Releases, Investor Presentations, Scientific Journals, Expert Interviews, and MarketsandMarkets Analysis

2.5 NEXT-GENERATION DATA STORAGE MARKET, BY STORAGE MEDIUM

FIGURE 7 NEXT-GENERATION DATA STORAGE MARKET FOR SSD TO GROW AT HIGHEST CAGR DURING FORECAST PERIOD



	SSD	HDD	Tape
2020	XX	XX	XX
2025	XX	XX	XX
CAGR (2020-2025)	XX%	XX%	XX%

Source: Press Releases, Investor Presentations, Scientific Journals, Expert Interviews, and MarketsandMarkets Analysis

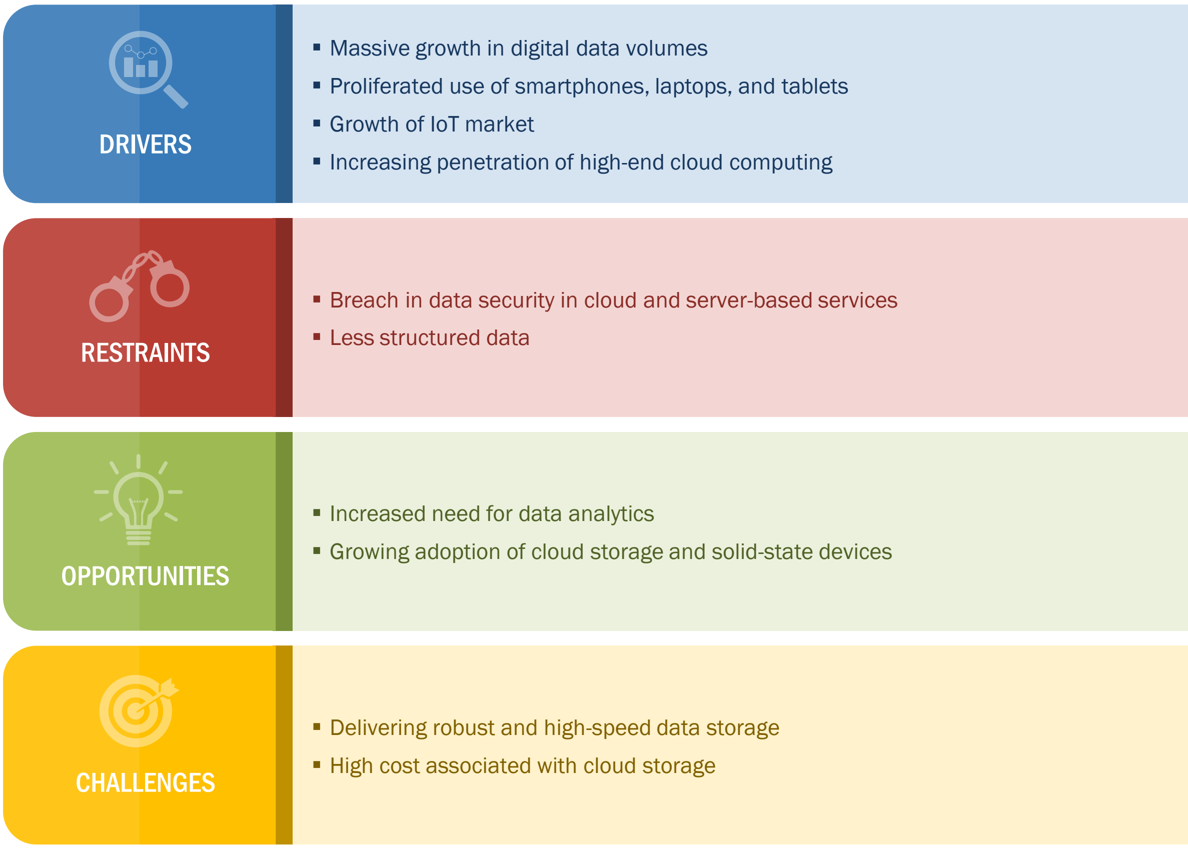
3 MARKET OVERVIEW

3.1 INTRODUCTION

Next-generation data storage technologies are adopted by organizations to store and access data from multiple, distributed, and connected resources. The volume of digital data being generated worldwide is growing at an exponential rate; this has driven the focus of enterprises on storing and managing these data. To store this vast data, companies are switching to technically advanced storage technologies. The storage market is gaining more traction as the enterprises need to access and store the data from multiple locations. This chapter explains the drivers, restraints, opportunities, and challenges for the next-generation data storage market.

3.2 MARKET DYNAMICS

FIGURE 8 MASSIVE SURGE IN VOLUME OF DIGITAL DATA IS MAJOR DRIVER FOR NEXT-GENERATION DATA STORAGE MARKET



Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

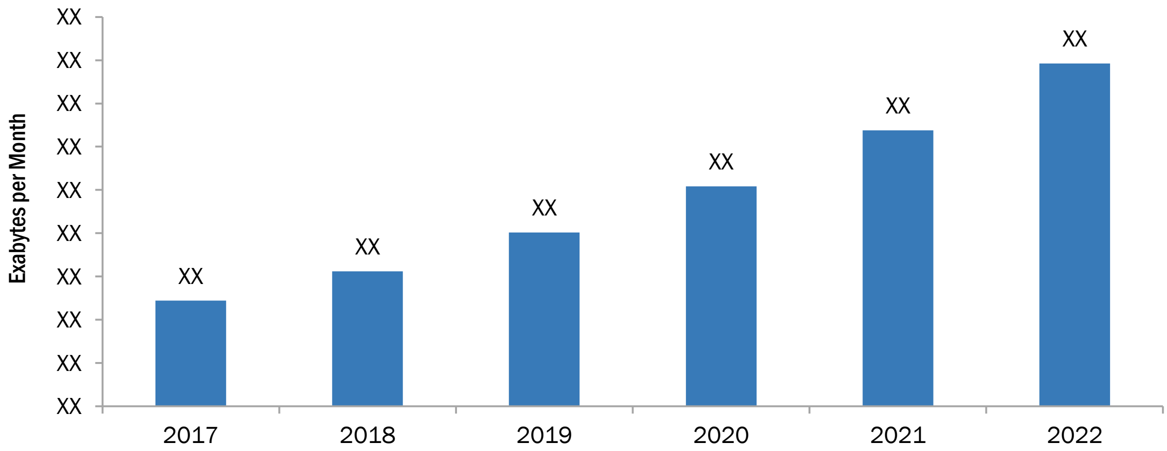
3.2.1 DRIVERS

3.2.1.1 Massive growth in digital data volumes

Digital data generation from various embedded systems, such as MP3 players, traffic lights, and MRI scanners, is growing at a breakneck pace. These massive data volumes are becoming a major challenge for data centers in managing their established practices. The major focus on retaining relevant data to get actionable insights makes the analysis of huge data volumes necessary. Moreover, most of the organizations have realized that the machine-generated data adds more value to their businesses than the data collected from customers and employees. With the Internet of Things (IoT) and connected devices gaining traction from the past few years, the volume of machine-generated data has increased tremendously. Several organizations are combining machine-generated data with manually collected data to get the best possible results. This has led to the generation of enormous volumes of digital data in recent times, and managing such huge volumes of data becomes a problem for many organizations. Storage software vendors provide software that can replicate and store data at different locations, so that organizations can retrieve those when required.

Increasing digitization across businesses has led to the creation of more content with over XX% data being in the form of documents, images, and audio/video files, contributing to the abundance of unstructured data and content. These factors have made it essential for businesses to explore efficient database applications as well as implement scalable content management and storage solutions across enterprises. The increase in internet protocol (IP) traffic is depicted in the below figure; it shows that the IP traffic would reach 396 EB per month by 2022. This increase in the data volume is attributed to emerging technologies such as IoT, robotics, connected living, connected cars, and Big Data.

FIGURE 9 IP TRAFFIC PER MONTH, 2017–2022



Source: Cisco Visual Networking Index: Forecast and Trends, 2017–2022

4 INDUSTRY TRENDS

4.1 INTRODUCTION

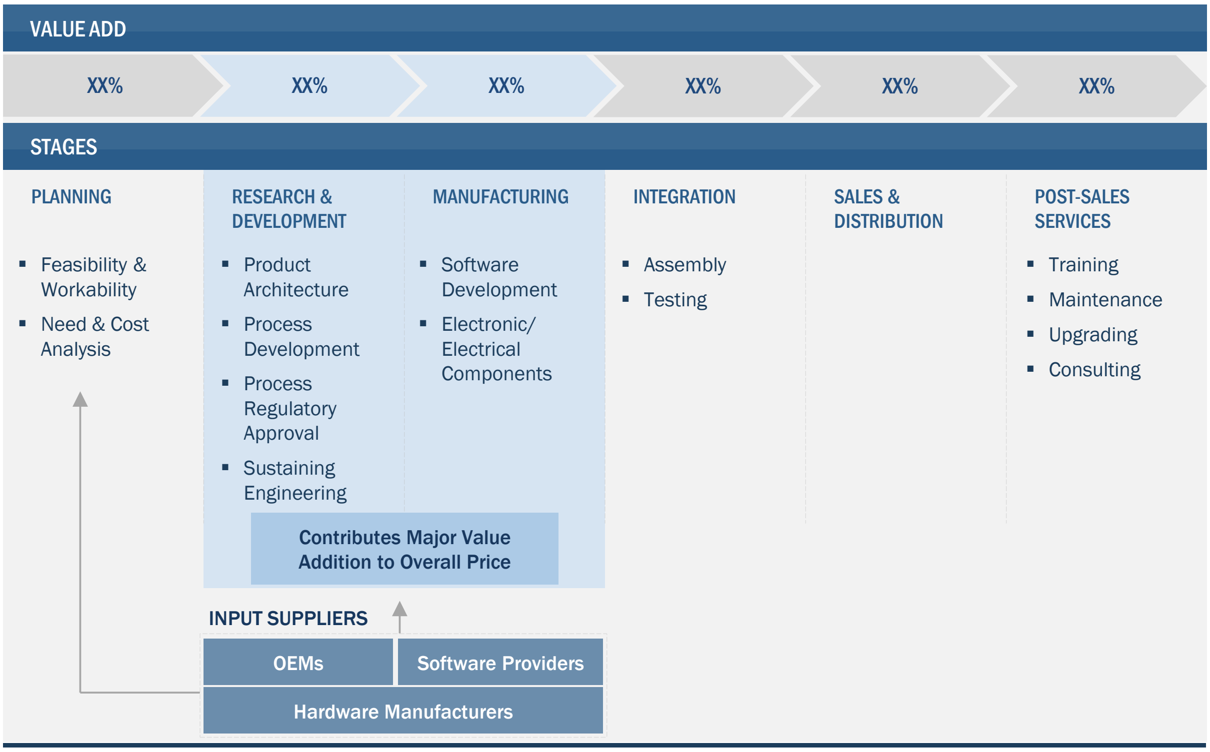
The next-generation data storage (NGDS) market is continuously evolving. The latest technologies and trends have transformed traditional data storage methods. They have revolutionized the data storage capabilities globally to address different challenges posed by large volumes of dense, complex, and unorganized data as well as the need for technologically advanced infrastructure for storing, fetching, and archiving data. The advancements in technologies such as the Internet of Things (IoT), artificial intelligence, and Big Data analytics have given rise to the need for new-generation data storage devices.

The demand for data storage devices is increasing in sectors such as banking, financial services, and insurance (BFSI), government, and media and entertainment, depending on the specific needs of users for data management. The development of new storage technologies, enabling saving on cost, time, and other resources, ensures high efficiency in storage operations.

This chapter provides a comprehensive analysis of the value chain of the next-generation data storage industry. It provides insights for understanding where the market is heading and the various factors impacting it.

4.2 VALUE CHAIN ANALYSIS

FIGURE 10 VALUE CHAIN: NEXT-GENERATION DATA STORAGE MARKET



Source: Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

The next-generation data storage is a customer-oriented market that offers user-specific solutions. The value chain of the next-generation data storage industry illustrates the collaboration between different vendors and customers for their offerings to perform effectively by addressing the ongoing business needs of data storage management. This would provide customers highly efficient, innovative, and application-focused solution.

This market includes multiple input suppliers for basic hardware technologies, which include magnetic storage, optical storage, and solid-state storage. The manufacturers and solution providers of data storage technology integrate new versions and security platforms continuously, and post-sales services play a major role in this value chain.

In the first stage of the value chain, manufacturers and service providers gather customers' business requirements for this market to plan and design infrastructure. Also, the study of risk and cost analysis, as well as feasibility and workability of products and integrated technologies for specific applications, needs to be carried out to ensure significant demand in the market. This stage is followed by R&D, which consists of product architecture, process development, process approval, and others.

In the manufacturing stage, application-specific hardware is developed as materials vary depending on the storage technology and framework used. Companies such as HPE Company, IBM, NetApp, and Dell are the leading players involved in the development and integration of software for data storage. Storage systems such as direct-attached storage (DAS), network-attached storage (NAS), storage area network (SAN), software-defined storage (SDS), cloud storage, and unified storage are selected according to the customer need. Supportive electronic devices are also assembled in this stage. The next stages of the value chain are distribution and post-sale services. Many companies in this market adopt the strategy of distributing their products through third-party suppliers for the global distribution of their offerings. New companies in this market, with limited geographic scope, have distribution contracts with global distributors. The end users of next-generation data storage are diversified worldwide; they include BFSI, consumer electronics, telecommunications and ITES, government, media and entertainment, and manufacturing sectors, among others. Others include transport and logistics, aerospace and defense, and automotive sectors. Regular maintenance, training, upgrading, and consulting are the key post-sale services that drive the inclusion of technology for commercial use.

5 NEXT-GENERATION DATA STORAGE MARKET, BY STORAGE SYSTEM

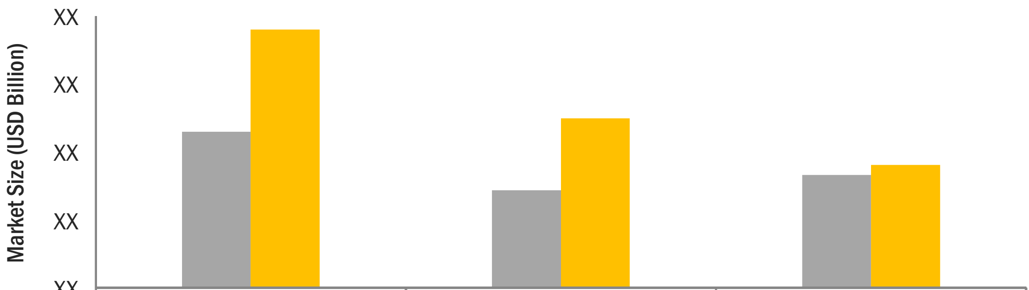
5.1 INTRODUCTION

Next-generation data storage technologies address storage-related challenges faced by multiple industries. Benefits of the latest data storage technologies include saving on time consumption, reduced hardware architecture, and virtualization and mobility. This chapter provides an analysis of the next-generation data storage market for different storage systems; the storage systems covered here are direct-attached storage (DAS), network-attached storage (NAS), and storage area network (SAN).

FIGURE 11 NEXT-GENERATION DATA STORAGE MARKET, BY STORAGE SYSTEM

Source: Annual Reports, IEEE Journals, Investor Presentations, Expert Interviews, Storage Networking Industry Association, AFCOM, Active Archive Alliance, Data Management Institute, Green Grid Consortium, and MarketsandMarkets Analysis

FIGURE 12 SAN TO HOLD LARGEST SIZE OF NEXT-GENERATION DATA STORAGE MARKET BY 2025



	Storage Area Network (SAN)	Network-attached Storage (NAS)	Direct-attached Storage (DAS)
2020	XX	XX	XX
2025	XX	XX	XX
CAGR (2020-2025)	XX%	XX%	XX%

Source: Annual Reports, IEEE Journals, Investor Presentations, Expert Interviews, Storage Networking Industry Association, AFCOM, Active Archive Alliance, Data Management Institute, Green Grid Consortium, and MarketsandMarkets Analysis

TABLE 1 NEXT-GENERATION DATA STORAGE MARKET, BY STORAGE SYSTEM, 2016–2019 (USD BILLION)

Storage System	2016	2017	2018	2019	CAGR (2016–2019)
Direct-attached Storage	XX	XX	XX	XX	XX%
Network-attached Storage	XX	XX	XX	XX	XX%
Storage Area Network	XX	XX	XX	XX	XX%
Total	**XX**	**XX**	**XX**	**XX**	**XX%**

Source: Annual Reports, IEEE Journals, Investor Presentations, Expert Interviews, Storage Networking Industry Association, AFCOM, Active Archive Alliance, Data Management Institute, Green Grid Consortium, and MarketsandMarkets Analysis

TABLE 2 NEXT-GENERATION DATA STORAGE MARKET, BY STORAGE SYSTEM, 2020–2025 (USD BILLION)

Storage System	2020	2021	2022	2023	2024	2025	CAGR (2020–2025)
Direct-attached Storage	XX	XX	XX	XX	XX	XX	XX%
Network-attached Storage	XX	XX	XX	XX	XX	XX	XX%
Storage Area Network	XX	XX	XX	XX	XX	XX	XX%
Total	**XX**	**XX**	**XX**	**XX**	**XX**	**XX**	**XX%**

Source: Annual Reports, IEEE Journals, Investor Presentations, Expert Interviews, Storage Networking Industry Association, AFCOM, Active Archive Alliance, Data Management Institute, Green Grid Consortium, and MarketsandMarkets Analysis

The COVID-19 pandemic and the subsequent lockdown caught businesses off-guard, forcing them to enable their employees to work from home. With employees of all types of enterprises working from home, NAS has emerged as an extremely powerful storage option for the employees connecting through VPNs. This has also compelled organizations to shift more of their applications and data to the cloud, and transition from a cloud-first to a cloud-only architecture. . The massive unstructured data, which is a file data generated by audio, video, and rich graphics, among others, is leading to the increased demand for innovative file storage products, which can scale bandwidth and performance to handle the increased data generation in a cost-effective manner. The demand for evolved NAS systems is increasing to cope with unstructured data growth. NAS is a dedicated storage device with multiple racks of storage media and is set up onto a dedicated network for storing the data. Scale-out is an improved technique of storage solution used in NAS devices, enabling the expansion of the capacity based on the requirement of end users. This is achieved with the help of clustered nodes; as a result, the cost of upgrading reduces. NAS devices are connected directly to the network and can give access to multiple authentic users. NAS storage devices offer cost-effective data storage solutions to organizations. Small businesses and medium-sized enterprises have started adopting cloud-based storage on NAS, owing to their ease and beneficial features. The demand for storage devices that are attached to a given network and data accessibility from any connected device for multiple authenticated users via a network are driving the growth of the market for NAS.

SAN storage facilitates the pooling of storage among data centers. SAN lead to easier centralization of storage management, which redistributes storage resources to other servers. Currently, large companies are the dominant adopters of SAN storage; however, the demand for SAN storage is expected to grow among the number of small and mid-size companies owing to its decreasing implementation costs. Additionally, SAN storage also contributes to the virtualization of data centers, which further leads to the high demand for this storage type. As an increasing number of companies have started utilizing cloud-based services to store their data with the use of virtual servers, companies aim to expand their storage capacities beyond existing infrastructural capabilities. Cloud technology creators widely use SAN technology due to its ability to connect large numbers of servers to storage devices.

5.2 DIRECT-ATTACHED STORAGE

5.2.1 DIRECT-ATTACHED STORAGE IS OLDEST AND CONVENIENT DATA STORAGE SYSTEMS

DAS is an individual storage system that is directly attached to a computer and can be accessed from a single system. Individual disk drives installed on a server for enterprise data management can also be the DAS systems. It is the oldest storage system and offers a convenient data storage solution. DAS is simpler and cheaper than any other storage system. It uses block storage technology. It uses hard disk drives (HDDs) and solid-state drives (SSDs) on a per-gigabyte basis for data storage.

TABLE 3 NEXT-GENERATION DATA STORAGE MARKET FOR DIRECT-ATTACHED STORAGE, BY END USER, 2016–2019 (USD BILLION)

End User	2016	2017	2018	2019	CAGR (2016–2019)
Enterprises	XX	XX	XX	XX	XX%
Cloud Service Providers	XX	XX	XX	XX	XX%
Telecom Companies	XX	XX	XX	XX	XX%
Government Bodies	XX	XX	XX	XX	XX%
Total	**XX**	**XX**	**XX**	**XX**	**XX%**

Source: Annual Reports, IEEE Journals, Investor Presentations, Expert Interviews, Storage Networking Industry Association, AFCOM, Active Archive Alliance, Data Management Institute, Green Grid Consortium, and MarketsandMarkets Analysis

TABLE 4 NEXT-GENERATION DATA STORAGE MARKET FOR DIRECT-ATTACHED STORAGE, BY END USER, 2020–2025 (USD BILLION)

End User	2020	2021	2022	2023	2024	2025	CAGR (2020–2025)
Enterprises	XX	XX	XX	XX	XX	XX	XX%
Cloud Service Providers	XX	XX	XX	XX	XX	XX	XX%
Telecom Companies	XX	XX	XX	XX	XX	XX	XX%
Government Bodies	XX	XX	XX	XX	XX	XX	XX%
Total	**XX**	**XX**	**XX**	**XX**	**XX**	**XX**	**XX%**

Source: Annual Reports, IEEE Journals, Investor Presentations, Expert Interviews, Storage Networking Industry Association, AFCOM, Active Archive Alliance, Data Management Institute, Green Grid Consortium, and MarketsandMarkets Analysis

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